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Factors affecting chatbots in banking services: the UTAUT2 and innovation resistance theory perspective

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Abstract

This paper aims to investigate the factors affecting the users' intention to use chatbots for banking service requirements. Based on the survey of 194 individuals, a partial least square and structural equation modelling (PLS–SEM) is employed to examine the factors affecting behavioural intention. Furthermore, a sub-sample analysis based on age (Gen Z and older age individuals) is performed to uncover the unique effects of various factors depending on the age groups. We find the positive effect of hedonic motivation, performance expectancy and social influence on behavioural intention, whereas the risk barrier has a negative effect on behavioural intention. Further sub-sample analysis reveals interesting results. Hedonic motivation's effect is observed only for Gen Z individuals, whereas the effect of social influence is observed only for the older age groups. The effects of performance expectancy and risk barrier are observed for both the sub-samples. The current study contributes to the extant literature in two ways. First, the findings contribute to the ongoing research on the application of Artificial Intelligence (AI) in general and chatbots in particular. Though there is growing research on chatbots in the last few years, there is a lack of empirical studies on chatbots for banking services specifically. Second, this study empirically validates the UTAUT2 and IRT theories, particularly the fusion of the two in explaining the innovation adoption.

Keywords: Artificial intelligence, Chatbots, Gen Z, PLS–SEM, Emerging economies

Introduction

In the digital age, we have entered what is often referred to as the “second machine age,” where artificial intelligence (AI) and robotics are integrated into everyday life, transforming the way we interact with technology (Brynjolfsson & McAfee, 2014). AI has become a cornerstone for creating exceptional customer experiences by enabling the analysis and interpretation of vast amounts of data that would otherwise be beyond human capacity (Oracle, 2017). Within this context, the financial sector has experienced a significant transformation, as AI-driven solutions have revolutionized how users and financial institutions engage with one another (Shaikh & Karjaluoto, 2015). One of the most prominent technological innovations reshaping the banking sector is the

use of AI-powered chatbots (Richad et al., 2019). As banks strive to meet the evolving needs of their customers, chatbots have emerged as a key tool for enhancing customer relationship management, providing personalized services, offering tailored financial solutions, and improving operational efficiency. The adoption of AI technologies, such as chatbots, enables financial institutions to not only streamline their processes but also address critical challenges, such as fraud detection, cost reduction, and customer retention.

The rise of chatbot technology in the banking sector has been notable, with banks increasingly deploying these systems to enhance service delivery. Chatbots are AI programs designed to facilitate human-like interactions through text or voice commands, automating customer service functions and improving overall customer experience. While rule-based chatbots were initially limited to predefined responses, the advent of machine learning has significantly expanded their capabilities, allowing them to learn from user interactions and provide more personalized solutions. Notable examples of AI-powered chatbots, such as Siri, Alexa, and ChatGPT, demonstrate the advanced capabilities of these systems, enabling them to deliver increasingly sophisticated and tailored experiences (Haugeland et al., 2022).

In the banking sector, the adoption of chatbot technology offers both consumers and banks a wealth of opportunities. However, the willingness of consumers to accept and engage with chatbots remains a critical factor in realizing the full potential of this technology. Understanding the drivers of consumer acceptance is crucial for banks to effectively implement chatbots and ensure their widespread adoption. While previous research has explored the use of AI chatbots across various industries, including aviation, luxury goods, and customer service, studies focusing specifically on the banking sector are still limited (Appendix 1). In India, the adoption of chatbots in banking services has gained momentum in recent years, with both private and public sector banks introducing their own chatbot systems. Notable examples include Eva by HDFC Bank, Keya by Kotak Mahindra Bank, and Uva by Union Bank of India (Appendix 2). Recent statistics indicate that 80% of Indian banks have integrated chatbots into their services (The Hindu, 2024, January 6). These chatbots are often designed with human-like names and personas to enhance user engagement and create more relatable experiences for customers.

While previous studies have explored the acceptance of mobile banking, internet banking, and other digital innovations in the financial sector, there is a gap in the literature regarding the adoption of chatbots in banking. This study aims to fill this gap by examining the factors that influence consumers' intention to use chatbots for banking services. Drawing on the Unified Theory of Acceptance and Use of Technology 2 (UTAUT2) and Innovation Resistance Theory (IRT), this research investigates both the facilitators and inhibitors that affect chatbot adoption in the banking sector. In addition, we conduct a sub-sample analysis by dividing respondents into two age groups—Gen Z and older individuals—to better understand the unique factors that influence chatbot acceptance across different demographic segments. By focusing on the Indian banking sector, this study provides new insights into the adoption of AI technologies in an emerging economy context, offering valuable implications for both researchers and practitioners in the field. This study makes two key contributions. First, it contributes

to the growing body of literature on AI applications, particularly chatbots, by focusing on their use in banking services—a relatively underexplored area. Second, it extends the application of the UTAUT2 and IRT frameworks, offering an empirical validation of these theories in the context of innovation adoption in the banking industry.

Theoretical background

We draw on UTAUT2 and IRT theories to explain the factors affecting the behavioural intention to use chatbot services for banking requirements. Appendix 3 shows that UTAUT2 and IRT have been widely used to explain the intention and adoption of innovation. While some empirical studies have used UTAUT2 and IRT separately to explain the intention and adoption of innovation, few recent studies have combined the two theories to explain the same phenomenon. Consistent with Sivathanu (2019), Migliore et al. (2022) and Khoo et al. (2024), we draw on these two theories to investigate the factors affecting the behavioural intention to use chatbot services.

UTAUT2

UTAUT2, an extension of the Unified Theory of Acceptance and Use of Technology (UTAUT), offers a comprehensive framework for understanding the adoption and usage of technology across different contexts (Venkatesh et al., 2012). It integrates elements from established theories, such as the theory of reasoned action, technology acceptance model, theory of planned behavior, and innovation diffusion theory to address the complexities of technology adoption. The model identifies key constructs influencing individuals' decisions, including performance expectancy, effort expectancy, social influence, facilitating conditions, hedonic motivation, price value, habit, and voluntariness. These constructs interact to predict users' intentions and behaviors regarding technology adoption, considering factors, such as perceived usefulness, ease of use, social pressures, and individual preferences. UTAUT2 is widely applied in research and practice to analyze technology acceptance in various domains, including chatbot adoption in different areas (Almahri et al., 2020; García de Blanes Sebastián et al., 2022; Paraskevi et al., 2023).

Innovation resistance theory

Various studies have researched into the obstacles hindering consumer adoption of innovations, often relying on the Innovation Resistance Theory (IRT) as a guiding framework (Kaur et al., 2020; Qasem & Hmoud, 2024). IRT suggests that consumers resist innovation due to its association with change, challenging established habits and beliefs, with barriers categorized into functional (usage, value, risk) and psychological (image, tradition) realms. Resistance, seen as a natural response to disruptive change, can be passive (conflicts with beliefs) or active (concerns about innovation's characteristics). Organizations must grasp these hints for successful market penetration. Some studies have employed IRT, combined with complementary theories to explore barriers to technological innovation adoption, highlighting its versatility and relevance in understanding consumer behavior toward innovation (Werth & Breitner, 2019; Zou et al., 2024).

Hypothesis development

Performance expectancy and behavioral intention

Performance Expectancy, a key factor in the Technology Acceptance Model (TAM), plays a crucial role in influencing Behavioral Intention to use technology. When users perceive that a particular technology will enhance their performance and productivity, they are more likely to express a positive intention to adopt and use it (Oliveira et al., 2016). This positive expectation stems from the belief that the technology will provide tangible benefits, such as efficiency improvements, time savings, or better task outcomes (Toufaily et al., 2021). Users tend to weigh the perceived advantages against potential challenges, and when the perceived performance gains outweigh any perceived drawbacks, it fosters a stronger intention to adopt the technology (Adapa et al., 2020). In essence, the perceived positive impact on performance becomes a driving force in shaping users' behavioral intentions to embrace and integrate the technology into their regular routines. Literature suggest that when users perceive chatbots as efficient tools for swiftly obtaining comprehensive banking information and services, it creates a positive attitude and contribute to their inclination to engage and to seamlessly integrate chatbots into their routine banking service interactions (Hari et al., 2022). Thus, we propose:

H1: Performance expectancy positively influence behavioral intention to adopt chatbot services.

Effort expectancy and behavioural intention

Effort Expectancy, a vital component of the Unified Theory of Acceptance and Use of Technology 2 (UTAUT 2), exerts a positive influence on behavioural intention to use technology (Nikolopoulou et al., 2021). When users anticipate that utilizing a particular technology will be straightforward, easy, and require minimal effort, it significantly enhances their intention to adopt the technology. Literature suggest that a perceived ease of use fosters a positive user experience, reducing resistance and creating a more favourable attitude toward the technology (Mullins & Cronan, 2021). Users are more likely to embrace and integrate the technology into their daily activities when they believe that interacting with it demands little cognitive or physical effort. Previous findings suggest that the positive perception of low effort expectancy acts as a catalyst, facilitating a smoother transition from intention to actual adoption of the technology (Rajak & Shaw, 2021). Thus, we propose,

H2: Effort expectancy positively influence behavioral intention to adopt chatbot services.

Social influence and behavioral intention

Within the realm of technology adoption, literature highlights the pivotal role of social influence in fostering a positive impact on behavioral intention (Gupta & Arora, 2020). When users perceive that their peers, colleagues, or social network endorse and support the use of a specific technology, it significantly enhances their intention to adopt it. The influence of social interactions, recommendations, and approval creates a sense of social norm and acceptance, reducing uncertainties and concerns related to the new technology

(Gupta & Arora, 2020). Individuals are more likely to embrace and integrate the technology into their routines when they observe others in their social circles endorsing its usage, as this social validation fosters a positive attitude and a stronger intention to adopt the technology (Kamal Basha et al., 2022). The social validation of chatbot usage is expected to play a crucial role in influencing users' future intentions regarding their adoption for banking services. Thus, we propose:

H3: Social influence positively influence behavioral intention to adopt chatbot services.

Facilitating conditions:

Previous findings suggest that facilitating conditions exert a positive influence on behavioral intention to adopt technology by addressing practical aspects of the adoption process (Ho et al., 2020).

When users perceive that the necessary infrastructure, technical support, and resources are readily available to support the use of a particular technology, it enhances their intention to adopt it (Rodríguez-Espíndola et al., 2022). The assurance of smooth integration and ongoing support creates a favorable environment for users, reducing barriers and uncertainties. According to Kamal et al. (2020) users are more inclined to adopt the technology when they feel confident that they have the necessary conditions and resources in place to facilitate its use effectively. Thus, facilitating conditions contribute significantly to fostering a positive intention to embrace and incorporate the technology into daily practices. Therefore, we propose:

H4: Facilitating conditions positively influence behavioral intention to adopt chatbot services.

Hedonic motivation:

Hedonic motivation in technology adoption refers to the pleasure or enjoyment individuals derive from using a particular technology. Literature suggest that when individuals perceive that using a specific technology will provide them with enjoyable and pleasurable experiences, it significantly enhances their intention to adopt it (Vishwakarma et al., 2020). The anticipation of fun, entertainment, or emotional satisfaction associated with the technology creates a positive attitude toward adoption. According to Siyal et al. (2020) users are more likely to embrace the technology when they expect it to fulfil their hedonic needs, adding an extra layer of motivation beyond utilitarian benefits. This emotional appeal not only enriches the overall user experience but also contributes to a more robust and positive intention to integrate the technology into their daily lives (Al-Azawei & Alowayr, 2020). When users perceive using chatbots as not only functional but also as a source of enjoyment, fun, and entertainment, it is likely to enhance their intention to incorporate chatbots into their routine banking interactions. Thus, we propose:

H5: Hedonic motivation positively influence behavioral intention to adopt chatbot services.

Traditional barrier and behavioural intention to use

Psychological resistance to innovation primarily stems from the cultural shift imposed on customers. According to van Klyton et al. (2021) when an innovation necessitates a departure from well-established traditions, it is met with resistance. The level of resistance tends to increase in proportion to the extent of deviation required. Literature suggests that traditional barriers play a pivotal role in shaping individuals' behavioral intention to adopt technology (Shanmugavel & Balakrishnan, 2023). Cultural norms, societal expectations, and long-standing habits create resistance to change, hindering the acceptance of new technological advancements. In addition, generational gaps in technology familiarity can also contribute to resistance (Chan & Lee, 2023). Older individuals may be less comfortable with or accustomed to using new technologies, leading to resistance. Thus, we propose:

H6: Tradition barrier negatively influence behavioral intention to adopt chatbot services.

Risk barrier

The Risk Barrier is a crucial determinant influencing behavioral intentions, as indicated by consistent findings in the literature of technology adoption (Ngoc et al., 2023). Perceived risks, encompassing financial, performance, and social dimensions, exert a significant influence on users' decisions to adopt technology. Literature also suggest that the fear of adverse health effects, high expenses, unproven performance, and potential social repercussion collectively contribute to a pervasive reluctance among individuals and organizations to adopt new technological advancements (Behera et al., 2023). Users may be hesitant to share sensitive information or engage in conversations with chatbots if they perceive potential risks to the confidentiality of their data. The extent to which chatbot interactions are perceived as risky directly influences users' willingness to adopt this technology. Thus, we propose:

H7: Risk barrier negatively influence behavioral intention to adopt chatbot services.

Image barrier

In the landscape of technology adoption, the "Image Barrier" emerges as a significant challenge, influenced by social and psychological factors. Users' willingness to embrace a technology is shaped by the perceived image attached to it; a positive image enhances adoption, while a negative one acts as a deterrent. Ram and Sheth (1989) note that these barriers can be triggered by factors, such as product category, country of origin, and brand. Mahmud et al., (2023) links image barriers to technology readiness, a mix of beliefs and emotions toward technology in general. In the digital realm, Ghosh (2024) highlights the significance of the Image barrier in resistance to adopting digital payment methods. Acknowledging that consumers' negative associations can lead to conflicts, addressing these perceptions becomes crucial for the adoption of emerging technologies, such as chatbots. Thus, we propose:

H8: Image barrier negatively influence behavioral intention to adopt chatbot services.

Effect of age and gender on the intention to use

Empirical evidence indicates that attitudes toward new technology vary significantly with age. Younger individuals are generally more adaptable and receptive to technological changes compared to older individuals (Ali et al., 2016). Several studies have shown that age influences the relationships between technological characteristics and the intention to adopt or actual adoption of technology. Natarajan et al. (2018) found that the effect of perceived ease of use on the intention to use mobile shopping applications is stronger among the younger age groups. Wang (2008), while studying acceptance of mobile learning applications, found that age moderates the effect of effort expectancy and social influence on mobile learning applications. Furthermore, research suggests that younger and older consumers differ in their perceptions of the risks associated with technology use. Age acts as a moderating factor in the relationship between perceived barriers and the adoption of new technologies (Chaouali & Souiden, 2019; Laukkanen et al., 2007). Consequently, age influences both the relationship between technological characteristics and the intention to use chatbot services, as well as the relationship between perceived barriers and the intention to adopt such services.

Gender differences significantly influence the intention to adopt and use technology. Tsourela and Roumeliotis (2015) found that women are more influenced by the perceived benefits of such services compared to men. Similarly, Nysveen et al. (2005) demonstrated that gender plays a critical role in shaping the intention to use mobile chat services. Specifically, women's intentions are predominantly driven by social norms and intrinsic motivators, such as enjoyment, whereas extrinsic factors, such as perceived usefulness more strongly influence men's intentions. Furthermore, research has shown that gender moderates the relationship between technological factors and both the intention to adopt or the actual adoption of new technologies (Faqih, 2016; Park et al., 2019).

Based on the variables discussed above, i.e., UTAUT2, IRT, and control variables and the intention to use chatbot services, we present the proposed research model in Fig. 1.

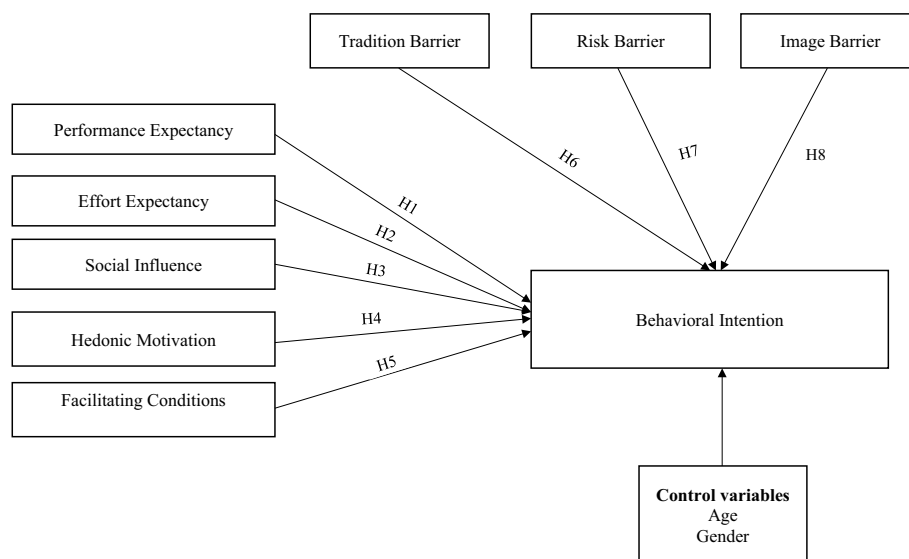


Fig. 1 Research model. Source: Authors

Research methodology

Survey instrument

The survey questionnaire was developed by reviewing the extant literature, thus ensuring the content validity of the survey instrument. We adapted the scales with slight modifications from the works of (Migliore et al., 2022), (Hasal et al., 2021) and (Gansser & Reich, 2021). All the constructs, except for two control variables, were measured on multiple items five-point Likert-type scale (anchored at 1 = “strongly disagree” and 5 = “strongly agree”). To measure the age, respondents were asked to fill in their age (in years). Gender is measured on a categorical scale, whereby they were given the options in terms of male, female or others.

Population sample, sample size, and survey administration

The study's target population consisted of Indian individuals aged 18 or older who hold bank accounts and have access to some level of chatbot services provided by their banks. A purposive sampling technique was employed, using the specified criteria to select participants for the survey. This approach was based on the assumption that the selected sample represents the population of interest and is well-suited to achieving the research objectives. The survey questionnaire was administered online. In total, an online questionnaire was sent to 250 respondents, out of which 194 complete responses were received. Thus, the response rate of the survey is 77.60%. As this study used partial least square-based structural equation modeling (PLS-SEM) for minimum sample size, we referred to both the ten times rule of thumb (Hair et al., 2011) and recent recommendations with regard to the use of power analysis. Hair et al. (2017) suggested the use of power analysis to calculate the minimum sample size. We used G*Power to perform power analysis, and it produced a minimum sample size of 172. Thus, it can be concluded that the 194 complete responses received in this study denote the sample adequacy. The sample size of 194 also meets the ten-time rule of thumb.

Common method bias

This study checks the common method bias using the Harman one-factor test, and it can be seen that the one factor derived from the principal component analysis explains only 27% of the total variance. Thus, it shows no issue of common method bias, as one factor explains less than 50% of the total variance (Podsakoff & Organ, 1986).

Data analysis techniques

This study used PLS-SEM for model measurement and structural model testing. Several studies have highlighted the usefulness and advantages of PLS-SEM in performing SEM (see Tiwari et al., 2023). While at the model measurement stage, factor loadings and internal consistency of the scales were checked. In addition, the convergent and discriminant validity of the latent constructs were examined during the

model measurement stage. The second stage in PLS–SEM involves model assessment, wherein the significance of the path coefficients was checked along with the tests for multicollinearity amongst the independent variables.

Analysis and results

Descriptive statistics

Table 1 shows the sample characteristics in terms of demography. It can be seen that the majority of the respondents belong to the less than 20 year group (50.52%), followed by 20–30 years (15.46%) and 30–40 years (15.46%). Meanwhile, in terms of gender, the majority of the respondents are male (69.01%) in comparison with their female counterparts (32.99%).

Model measurement

Model measurement

Table 2 shows the reliability and validity measures. The outer factor loadings of all the items of the latent constructs are greater than 0.7, except for IB2, which is retained to avoid a single-item scale. The study used both Cronbach's alpha and composite reliability (CR) to examine the internal consistency of the scales. The Cronbach's alpha and CR of all the constructs are greater than 0.7, thus indicating the internal consistency of all the constructs. To measure the convergent validity, we used Average variance explained (AVE). According to (Sarstedt et al., 2014), "*convergent validity measures the extent to which a construct converges in its indicators by explaining the items' variance*". The AVE of all the constructs is greater than 0.5, showing that, on average, each of the constructs explains more than 50% variance of its own items. Thus, confirming the convergent validity.

We measured the discriminant validity of the constructs using both the Fornell–Larcker criterion and the heterotrait–monotrait (HT–MT) ratio. Discriminant validity indicates the extent to which a latent construct is empirically distinct from all the other constructs in the path model (Sarstedt et al., 2014). As per the Fornell–Larcker criterion, the square root of the AVE of the construct needs to be greater than its highest correlation with all the other constructs in the path model. Table 3 shows that the square root of AVEs is greater than its correlations with all the other constructs. Furthermore,

Table 1 Sample profiles

Characteristics	Frequency	Percentage
Age of the respondents (in years)		
Less than 20	98	50.52
20–30	30	15.46
30–40	30	15.46
40–50	25	12.89
50–60	06	3.09
Above 60	05	2.58
Gender		
Male	130	69.01
Female	64	32.99

Table 2 Details of the validity measures

Items description		Mean	Standard deviation	Standardized factor loadings	Cronbach's alpha	Composite reliability	AVE
Behavioural Intention (BI)							
BI1	In the future, I intend to use chatbots for my banking service requirements	3.464	1.149	0.894	0.928	0.949	0.823
BI2	I plan to use chatbots for my banking service requirements in the coming months	3.180	1.232	0.909			
BI3	I will try to use chatbots for my banking service requirements on a regular basis	3.077	1.192	0.903			
BI4	I will recommend others to use chatbots for banking service requirements	3.201	1.195	0.923			
Effort Expectancy (EE)							
EE1	Learning how to use a chatbot for banking is easy for me	3.763	1.013	0.797	0.848	0.898	0.687
EE2	My interaction with a chatbot is clear and understandable	3.505	1.042	0.843			
EE3	I find chatbots easy to use	3.799	0.945	0.847			
EE4	It is easy for me to become skilful at using a chatbot	3.686	0.979	0.827			
Facilitating Conditions (FC)							
FC1	I have the resources necessary to use a chatbot	3.918	1.017	0.873	0.750	0.858	0.671
FC2	I have the knowledge necessary to use a chatbot	3.825	1.045	0.857			
FC3	I can get help from others when I have difficulties using a chatbot	3.696	1.023	0.718			
Hedonic Motivation (HM)							
HM1	Using a chatbot for banking is fun	3.227	1.045	0.916	0.916	0.947	0.857
HM2	Using a chatbot for banking is enjoyable	3.175	1.131	0.956			
HM3	Using a chatbot for banking is very entertaining	2.995	1.119	0.905			
Image Barrier (IB)							
IB1	Integrated logistics systems	2.608	1.080	1.000	0.720	0.777	0.654
IB2	Cost control capabilities	2.619	1.056	0.555			
Performance Expectancy (PE)							

Items description		Mean	Standard deviation	Standardized factor loadings	Cronbach's alpha	Composite AVE reliability	
PE1	A chatbot is a useful method to avail banking information/ services or query	3.758	0.968	0.862	0.873	0.914	0.727
PE2	Chatbot enables me to avail banking information/services quickly	3.691	1.024	0.900			
PE3	Using a chatbot helps me avail banking information/services more effectively	3.459	1.099	0.891			
PE4	Using a chatbot allows me to save time	3.747	1.071	0.748			
Risk Barrier (RB)							
RB1	<i>I am concerned with cyber security while interacting with the chatbot^a</i>				0.907	0.955	0.914
RB2	I fear that my financial and personal information will be compromised from my interactions with the chatbot	3.423	1.111	0.943			
RB3	I fear that I am making myself vulnerable to cyber-attacks while using the chatbot	3.345	1.144	0.968			
RB4	<i>I am concerned that the chatbot will give the wrong information^a</i>						
RB5	<i>I am concerned that the bank will use the information from my chatbot interaction for other purposes^a</i>						
Social Influence (SI)							
SI1	People who influence my behaviour think that I should use a chatbot for banking	2.995	1.023	0.870	0.875	0.923	0.800
SI2	People who are important to me think that I should use a chatbot	2.985	1.096	0.914			
SI3	People whose opinions I value prefer that I use a chatbot	3.026	1.072	0.898			
Tradition barrier (TB)							

Table 2 (continued)

Items description		Mean	Standard deviation	Standardized factor loadings	Cronbach's alpha	Composite AVE reliability	
TB1	I prefer chatting with customer executives for my banking query	3.675	1.037	0.931	0.825	0.878	0.708
TB2	<i>I prefer chatting with a financial advisor for my banking service requirements^a</i>						
TB3	I think that customer executive gives a better feeling to resolving banking query	3.804	1.001	0.878			
TB4	I think that a financial advisor gives a better feeling to addressing banking service requirements	3.825	0.963	0.698			

^a Items not included in the PLS–SEM model due to unsatisfactory factor loadings

Table 3 Fornell–Larcker criterion

	AG	BI	EE	FC	GE	HM	IB	PE	RB	SI	TB
AG	1.000										
BI	0.063	0.907									
EE	– 0.045	0.497	0.829								
FC	– 0.104	0.336	0.644	0.819							
GE	0.283	– 0.072	– 0.034	0.085	1.000						
HM	0.030	0.657	0.489	0.326	0.013	0.926					
IB	– 0.126	0.065	– 0.122	– 0.079	– 0.042	0.023	0.809				
PE	– 0.032	0.730	0.647	0.486	– 0.065	0.657	– 0.002	0.852			
RB	– 0.050	– 0.086	– 0.011	0.061	– 0.113	– 0.056	0.354	– 0.003	0.956		
SI	0.049	0.560	0.515	0.420	0.032	0.541	0.091	0.522	0.015	0.894	
TB	– 0.089	– 0.103	0.114	0.239	0.086	– 0.065	0.041	– 0.036	0.043	0.005	0.842

AG Age, **BI** Behavioural Intention, **EE** Effort Expectancy (EE), **GE** Gender, **HM** Hedonic Motivation, **IB** Image Barrier, **PE** Performance Expectancy, **RB** Risk Barrier, **SI** Social Influence, **TB** Traditional Barrier

Table 4 Heterotrait–monotrait ratio (HTMT) matrix

	AG	BI	EE	FC	GE	HM	IB	PE	RB	SI	TB
AG											
BI	0.064										
EE	0.047	0.558									
FC	0.126	0.400	0.813								
GE	0.283	0.097	0.042	0.101							
HM	0.049	0.709	0.547	0.392	0.043						
IB	0.150	0.087	0.238	0.171	0.031	0.062					
PE	0.047	0.808	0.756	0.595	0.070	0.731	0.097				
RB	0.056	0.092	0.067	0.108	0.122	0.059	0.424	0.059			
SI	0.053	0.620	0.592	0.517	0.035	0.600	0.128	0.595	0.057		
TB	0.103	0.103	0.194	0.341	0.086	0.072	0.069	0.086	0.077	0.063	

Table 5 Path coefficients and f^2 effect sizes

Path	Full sample		Sub-sample 1		Sub-sample 2	
	Coefficients	f^2 size effect	Coefficients	f^2 size effect	Coefficients	f^2 size effect
AG→BI	0.085**	0.017				
EE→BI	0.004	0.000	− 0.091	0.010	0.105	0.015
FC→BI	− 0.010	0.000	0.078	0.007	− 0.158	0.054
GE→BI	− 0.161	0.014	− 0.431***	0.111	0.525***	0.128
HM→BI	0.232**	0.074	0.351***	0.219	− 0.050	0.003
IB→BI	0.092	0.018	0.054	0.006	0.090	0.030
PE→BI	0.483***	0.260	0.471***	0.315	0.595***	0.356
RB→BI	− 0.108*	0.027	− 0.143*	0.039	− 0.154*	0.087
SI→BI	0.177**	0.050	0.106	0.019	0.311***	0.197
TB→BI	− 0.055	0.007	− 0.093	0.020	− 0.057	0.012

***, ** and * refers to significance at 1%, 5% and 10%, respectively

the HTMT ratios of all the constructs are less than 0.90 (Table 4). Therefore, both the Fornell–Larcker criterion and HTMT ratio confirm the discriminant validity of the constructs.

Structural model assessment

At first, we checked for the multicollinearities amongst the latent constructs using the Variance Inflation Factor (VIF). We observed that the lowest VIF was 1.121 and the highest VIF was 2.469, thus showing no issue of multicollinearity amongst the independent variables in the path model (Hair et al., 2017).

The structural model testing was performed using the bootstrapping method with 5000 samples, as suggested by (Hair Jr et al., 2016). Overall, three structural models were performed, i.e., one full sample and two sub-samples. Sub-sample models were created to capture the role of generations in influencing the behavioural intention toward chatbots. Sub-sample 1 comprises respondents belonging to Gen Z (18 years to less than or equal to 26 years). Whereas, sub-sample 2 comprises respondents belonging to other generations older than Gen Z (above 26 years).¹ The results of the analysis are shown in Table 5.

For the full sample, all the independent variables explain 63.4% ($R^2=0.634$) of the variation in the behavioural intention toward chatbots. Analysing the paths, it can be seen that the paths PE→BI ($\beta=0.433$, p value <0.01), SI→BI ($\beta=0.177$, p value <0.05), HM→BI ($\beta=0.232$, p value <0.05) and RB→BI ($\beta=-1.08$, p value <0.10) are significant. While, performance expectancy, social influence and hedonic motivation and positively affect behavioural intention, risk barrier has a negative effect. Therefore, our results support the hypotheses H1, H3, H4 and H7, respectively. The results also show the positive and significant effect of age on behavioural intention ($\beta=0.085$, p value <0.05). Concerning the f^2 size effect, all the independent variables exhibited a significant effect on behavioural intention as indicated by t values and p values. While the PE→BI ($f^2=0.260$)

¹ Gen Zs are the people born between 1997–2012, which makes their age range to be 11–26 in 2023. In our sample the lowest age was 18 years. Thus, in sub-sample 1 (Gen Z group) we have respondents between the age 18–26 years.

exhibited a moderate effect size, all the other paths, i.e., AG→BI ($f^2=0.017$), HM→BI ($f^2=0.074$), RB→BI ($f^2=0.027$) and SI→BI ($f^2=0.050$) exhibited weak effect sizes.

We created two sub-sample models based on the generations. Sub-sample model 1 concerns the Gen Zs¹, and Sub-sample model 2 concerns other older generations. Regarding Sub-sample model 1, the independent variables explain 63.6% ($R^2=0.636$) of the variation in the behavioural intention toward chatbots. Analysing the paths, it can be seen that the paths PE→BI ($\beta=0.471$, p value < 0.01), HM→BI ($\beta=0.351$, p value < 0.01), and RB→BI ($\beta=-0.143$, p value < 0.10). While performance expectancy and hedonic motivation positively affected behavioural intention, the risk barrier showed a negative effect. Therefore, these results for the Gen Zs support the hypotheses H1, H4 and H7, respectively. The control variable, gender, also displayed a negative and significant effect for this sub-sample ($\beta=-0.431$, p value < 0.01). With regard to the f^2 effect size, moderate effect sizes are observed for the paths HM→BI ($f^2=0.219$) and PE→BI ($f^2=0.315$). Meanwhile, weak effect sizes are observed for the paths GE→BI ($f^2=0.111$) and RB→BI ($f^2=0.039$).

For the sub-sample 2 model, the $R^2=0.762$, thus indicating the 76.2% explanatory power of the model. It can be seen that the paths PE→BI ($\beta=0.595$, p value < 0.01), PE→BI ($\beta=0.311$, p value < 0.01) and RB→BI ($\beta=-0.154$, p value < 0.10). While performance expectancy and social influence showed a positive effect on behavioural intention, the risk barrier displayed a negative effect. Therefore, these results for this age group support the hypotheses H1, H3 and H7, respectively. The control variable, gender, also displayed a positive and significant effect for this sub-sample ($\beta=-0.525$, p value < 0.01). With regard to the f^2 effect size, PE→BI ($f^2=0.356$) exhibited a strong effect size. Whereas SI→BI ($f^2=0.197$) showed a moderate effect size. Finally, weak effect sizes are observed for the paths GE→BI ($f^2=0.128$) and RB→BI ($f^2=0.087$).

Discussion

The results of this study showed the positive influence of performance expectancy on the intention to use chatbot services in the banking sector, thus supporting hypothesis H1. This indicates that individuals expect the chatbot services to offer benefits for their banking requirements. Individuals may want chatbots to improve their productivity and performance. They may expect chatbots to provide time savings and better services in resolving queries and banking requirements than calling the banking executive or visiting a bank branch. Our result is consistent with the previous studies that showed that individuals look for performance improvement while using innovative technologies (Huang et al., 2021; Mohd Rahim et al., 2022; Ragheb et al., 2022; Sitthipon et al., 2022; Terblanche & Kidd, 2022).

We also found the positive influence of hedonic motivation on the intention to use chatbot services. It shows that individuals look for enjoyment and pleasurable experiences while using chatbots. Our findings are in line with the earlier studies on other innovative technologies (Gupta & Arora, 2020; Liébana-Cabanillas et al., 2021). The analysis of sub-samples reveals interesting results. It shows that seeking enjoyment and pleasurable experiences is prevalent in Gen-Zs, whereas individuals above 26 years do not seek such experiences in chatbot services. This is consistent with the findings of (Liébana-Cabanillas et al., 2021).

Social influence has also been found to positively influence the intention to use chatbot services. Seeing friends, family members, colleagues, etc., using chatbot services for banking needs is likely to influence the individual's intention to use chatbots. Social interactions, recommendations and approvals are likely to enhance the individual's intention to use chatbot services. This is line with the previous findings by (Ragheb et al., 2022), (Terblanche & Kidd, 2022). Sub-sample analysis shows exciting findings, whereby the effect of SI is prevalent in the older age groups (above 26 years) as compared to the Gen Zs. This shows that older age group individuals are likely to develop intention to use chatbot services when they see others using such innovative technologies. Previous studies have shown that older age groups are more hesitant to use innovative technologies unless they observe others using such technologies (Liébana-Cabanillas et al., 2014; Van De Watering, 2005). For this age group, social interactions, recommendations and approvals from other individuals play a crucial role in developing their intention to use any innovative technology. Encouraging word of mouth can be used to persuade individuals belonging to older age groups (Kaur et al., 2020).

Out of the UTAUT2 variables, we found effort expectancy and facilitating conditions to be insignificant in affecting intention to use chatbot services. The result of effort expectancy being insignificant is similar to the findings by (García de Blanes Sebastián et al., 2022; Joshi, 2021). One of the plausible reasons could be that the individuals have not properly explored or understood the full capabilities of chatbots or the individuals are indifferent to the ease or convenience of using chatbots for banking services. We also found facilitating conditions to be insignificant in affecting the intention to use chatbot services. Previous studies have also found facilitating conditions insignificant (García de Blanes Sebastián et al., 2022).

Only the risk barrier significantly influences the behavioural intention among the IRT variables. We found that the risk barrier is negatively associated with behavioural intention. As individuals' perception of risk in using chatbot services increases, they are less likely to use them for banking services. Privacy, confidentiality, and security are important parameters individuals look at when intending to use new innovations, particularly chatbots for banking services. As individuals perceive risks to their privacy, confidentiality, and security, their trust gets affected, and hence, they will be less willing to use such services (Terblanche & Kidd, 2022).

Image barrier is found to be insignificant in affecting behavioural intention. This finding is similar to the earlier studies (Kaur et al., 2020; Migliore et al., 2022). Similar to (Migliore et al., 2022), we argue that, when the level of technological orientation is high among individuals, the image barrier is less likely to affect the behavioural intention to use chatbots. We found tradition barrier to be insignificant in affecting behavioural intention. This is similar to the earlier findings by (Kaur et al., 2020). The probable explanation could be that the respondents do not realise a large departure or hindrance from using traditional channels to chatbots in meeting banking requirements. Another plausible reason could be our respondents are quite familiar with this kind of technology, especially with the advent of ChatGPT.

Conclusion

Artificial intelligence and chatbots have gained much attention in the last few years, both in academia and in practice. Chatbots are being introduced to assist and deliver services in different sectors, and the banking sector is one of them. Several banks in India, be it public sector banks or the private banks, have introduced their own chatbots to assist customers. This study focused on investigating the individual's intention to use chatbots for their banking service requirements. The results showed the positive influence of hedonic motivation, performance expectancy and social influence on behavioural intention to use chatbots. On the other hand, the risk barrier has a negative effect on behavioural intention. The sub-sample analysis revealed interesting insights concerning hedonic motivation and social influence. It showed that seeking enjoyment and pleasurable experiences is prevalent in Gen-Zs, whereas individuals above 26 years do not seek such experiences in chatbot services. In addition, the result showed that social influence is prevalent in the older age groups (above 26 years) compared to the Gen Zs, indicating that older age group individuals are likely to develop the intention to use chatbot services when they see others using such innovative technologies.

Theoretical contribution

The current study contributes to the extant literature in two ways. First, the findings contribute to the ongoing research on the application of artificial intelligence in general and chatbots in particular. Though, there is growing research on chatbots in the last few years, there is a lack of empirical studies on chatbots for banking services. Second, this study empirically validates the UTAUT2 and IRT theories, particularly the fusion of the two to explain the innovation adoption.

Managerial implications

The findings of this study offer several actionable insights for bank managers looking to enhance customer engagement and service through chatbots. First, emphasizing the performance benefits of chatbots—such as time savings, quick query resolution, and overall efficiency—can attract more users, as customers value these improvements. Banks should also focus on making chatbot interactions enjoyable and user-friendly to appeal particularly to younger customers (Gen-Z) who seek pleasurable digital experiences. In addition, highlighting positive social proof and endorsements from peers or trusted sources can encourage older customers to adopt chatbot services. Finally, addressing security and privacy concerns transparently will be crucial in mitigating perceived risks and building trust, ensuring more customers are comfortable using chatbots for their banking needs.

Limitations and future research directions

This study possesses a few limitations. It examined the individual's intention to use chatbots for banking services; however, actual adoption could be included.

Mixed-method designs, such as explanatory mixed-method design, can provide a rich understanding of the individual's intention and adoption of using chatbot services. Future research may explore moderating variables that enhance the UTAUT2 variables or dampen the negative effect of IRT variables on user intention.

Appendix 1 Empirical studies on chatbots

Sr no	Source	Country	Sector/focus area	Theory	Sample size	Analysis techniques	Results of the study
1	Chung et al., (2020)	South Korea	Luxury	–	157	Exploratory and confirmatory factor analysis using Amos	The study results that chatbots e-services gives an assistance engaging and interactive customer or brand services encounters
2	Wiliam et al., (2019)	Indonesia	Financial services	–	–	Neuro research with exploratory research stages	This study found that e-services chatbots can increase the banking customers satisfactions in Indonesia
3	Eren (2021)	Turkey	Financial services	Expectations confirmation theory	240	Structural equation modelling (SEM)	Perceived trust, perceived performance and corporate reputation impact the satisfaction of the consumer about the usages of the chatbots
4	Yun and Park (2022)	–	–	SERVQUAL theory	380	Exploratory and confirmatory factor analysis using Amos	The reliability and assurance of chatbots positively influence consumer satisfactions for both the groups. Interactivity and empathy positively influence on the consumer satisfaction only with emotion words and chatbots
5	Nguyen et al., (2021)	Vietnam	Financial services	DeLone and McLean's information systems success (D&M ISS) model and the expectation confirmation model (ECM)	359	Exploratory and confirmatory factor analysis using Amos	The user's continuance intentions toward the chatbot services banks were impacted by the trust, perceived usefulness and satisfactions and confirmation expectations were impacted on the 3 drivers of continuance intentions on various ways

Sr no	Source	Country	Sector/focus area	Theory	Sample size	Analysis techniques	Results of the study
6	Li et al., (2021)	China	Financial services	Post acceptance model of Information system	295	PLS–SEM	Among all the quality dimensions of the chatbot, i.e., understandability, assurance, reliability and interactivity are positively linked with the confirmations
7	Waarden et al. – (2020)	–	Airlines industry	SERVQUAL theory	146	SEM and mediation analysis using software-R,	Perceived usefulness and reliability are the main criteria that impact an intention to re-use the chatbot
8	Jang et al., (2021)	Korea	Financial services	Social representation theory	–	Content analysis, core periphery analysis of social representation theory	The findings of this study give a comprehensive and detailed view of the chatbot services in the Korean Financial industry
9	Alt et al., (2021)	Romania	Financial services	Technology acceptance model	287	PLS–SEM	The study highlights the importance of the perceived usefulness and perceived compatibility in the acceptance of the banking chatbot technology. Service awareness has an impact on the privacy risk, perceived ease of use and in directly impact the intention of usage of banking chatbots by perceived usefulness
10	Chen et al. (2023)	–	General organization	Trust transfer theory	459	PLS–SEM	The current study finds that AI chatbot services quality impacts the customer loyalty by cognitive trust, perceived values, satisfaction and affective trust
11	Silva et al., (2023)	Portuguese	–	Technology acceptance model	201	PLS–SEM	The findings reveal that perceived ease of use, perceived usefulness and trust have a positive influence on the attitude toward chatbot using
12	Abdallah et al. (2023)	Kuwait	Financial services	Technology acceptance model	385	PLS–SEM	It was found that there is positive association between AI chatbot quality and adoption of chatbot AI

Sr no	Source	Country	Sector/focus area	Theory	Sample size	Analysis techniques	Results of the study
13	Kuberkar and Singhal (2020)	India	Public transportation	UTAUT	463	PLS–SEM	This study empirically validates the intention to use AI chatbot for the public transportation with special reference to the big developing country like India
14	Mulyono and Sfenrianto (2022)	Indonesia	Financial services	–	100	PLS–SEM	The study found that service quality, system quality, IS positively impact the consumer satisfaction with special reference to Indonesian banking chatbot during COVID-19 pandemic
16	Hsiao and Chen (2022)	Taiwan	Food	Theory of reasoned action	111	PLS–SEM	The current study found that anthropomorphism and service quality-like problem solving, act as a predictor for the satisfaction and trust, where satisfaction has a strongly influence on the intentions of the users
17	Hsu and Lin (2023)	Taiwan	–	–	219	PLS–SEM	The study found that AI chatbot conversational quality and AI chatbot service quality has impacted the user satisfactions and loyalty of the users toward the chatbot services
17	Bouhia et al., (2022)	Canada	Financial services	Trait theory	430	PLS–SEM	The results found that privacy concerns emerged from feeling of creepiness, perceived risk and the privacy needs

Appendix 2 Chatbots of illustrative Indian banks

Name of the Bank	Sector	Name of the Chatbot	Year of introduction
HDFC Bank	Private sector	Eva	2017
Kotak Mahindra Bank	Private sector	Keya	2018
ICICI Bank	Private sector	iPal	2017
IndusInd	Private sector	Indus Assist	2018

Name of the Bank	Sector	Name of the Chatbot	Year of introduction
Axis Bank	Private sector	Aha	2018
State Bank of India	Public sector	No	No
Canara Bank	Public sector	Mr. CanBDunn	–
Union Bank of India	Public sector	Uva	2023
Bank of India	Public sector	Seva	–
Bank of Baroda	Public sector	Adi	2020

Appendix 3 Empirical studies using UTAUT2, IRT and UTAUT2 + IRT

Source	Theories used	Innovation/contribution	Sector	Type of study	Analysis technique	Key findings
Chakraborty et al., (2022)	IRT	Understanding user behavior and decision-making in fitness apps	Health and fitness	Quantitative research	Descriptive analysis	Users transitioning from traditional to app-based fitness methods enjoy new experiences; functional barriers do not hinder adoption; users seek health and fitness information from multiple sources; high health consciousness correlates with increased fitness activity
Kumar et al., (2024)	IRT	Extends IRT with social dependency, lack of facilitating conditions, and digital self-efficacy	Tourism and hospitality	Quantitative research	PLS–SEM	Significant predictors of discontinuance intentions include usage barrier, values barrier, risk barrier, lack of facilitating conditions, and digital self-efficacy; insights for tourism stakeholders and app developers on potential setbacks hindering hotel booking app usage
Musyaffi et al., (2022)	IRT	Adoption of digital payments by traditional clothing market traders during COVID-19	Traditional clothing market	Qualitative research	Thematic analysis	Indonesian traditional clothing traders face barriers to digital payments, prefer cash for simplicity and immediacy, and have concerns about data security and technical issues. Trust in well-known providers aids adoption, and digital proof reduces the need for external approval

Source	Theories used	Innovation/contribution	Sector	Type of study	Analysis technique	Key findings
Kumar et al., (2023)	IRT	Examining barriers that lead to customer disengagement in online communities	Online communities	Empirical study considering 301 responses	Hypothesis testing, mediation analysis, moderation analysis	Performance barriers, information overload, and social recognition barriers significantly increase disengagement intentions. Negative anticipated emotions mediate the relationship between these barriers and disengagement. Time spent in the community influences the impact of barriers on disengagement intentions
Jin et al., (2022)	IRT	Adoption of regenerative farming practices among farmers in Pakistan	Agriculture	Cross-sectional study considering 863 farmers	Descriptive analysis, correlation analysis, hypothesis testing	Communication campaigns effectively reduce resistance to adopting regenerative farming practices. Usage, value, risk, and traditional barriers are addressed; image barriers are not significantly impacted. Targeted communication strategies are crucial for promoting sustainable agricultural innovations
Macedo (2017)	UTAUT2	Technology acceptance among older adults	ICT	Empirical study considering 278 responses	Variance-based structural equation modelling (PLS)	UTAUT2 is valid for explaining older adults' behavioral intentions and ICT use. Predictors include behavioral intention, habit, performance expectancy, effort expectancy, social influence, facilitating conditions, and hedonic motivation
Palas et al., (2022)	UTAUT2	mHealth services adoption by elderlies in Bangladesh	mHealth services	Empirical study	Structural equations modelling (SEM), fsQCA	Social Influence, Hedonic Motivation, Price Value, Habit, and Service Quality significantly impact the intention to adopt mHealth services. Quality of Life does not significantly affect Behavioral Intention. fsQCA results show varying intensities of these factors contribute to different levels of mHealth adoption and use

Source	Theories used	Innovation/contribution	Sector	Type of study	Analysis technique	Key findings
Tavares et al., (2018)	UTAUT2	Adoption of EHR portals by patients	Healthcare	Empirical study considering 386 responses	Regression analysis	Performance expectancy, effort expectancy, social influence, and habit drive patients' intention to adopt EHR portals. Habit and behavioral intention significantly influence actual use. The model explains 52% of the variance in behavioral intention and 31% of the variance in technology use
Hanif et al., (2022)	UTAUT2	Determinants of mobile shopping intentions among young consumers in Pakistan	Mobile Shopping	Empirical study	Structural equation modelling (SEM)	Structural assurance, perceived risk, trust, and UTAUT constructs significantly influence mobile shopping intentions. Structural assurance and prior shopping experience moderate the effects of perceived risk and trust. Significant gender differences exist in mobile shopping behavior
Duarte and Pinho (2019)	UTAUT2	Factors influencing mobile health (mHealth) adoption	Mobile Health (mHealth)	Mixed Methods Study	PLS-SEM, fsQCA	Performance expectancy, hedonic motivation, and habit significantly predict mHealth adoption. Combining factors through fsQCA identifies additional sufficient conditions for mHealth adoption, suggesting multiple pathways (equifinality) exist
Migliore et al., (2022)	UTAUT2, IRT	Adoption gap in mobile payment systems between Italy and China	Mobile payment systems	Empirical study through a web-based, self-administered survey	Structural equation modelling (SEM), multi-group analysis	Performance Expectancy significantly influences mobile payment adoption in both countries, while social influence is significant only in Italy. Facilitating Conditions are important in China but not in Italy. Hedonic Motivation and Effort Expectancy do not significantly impact adoption. Cultural dimensions moderate the relationship between Social Influence and Behavioral Intention

Source	Theories used	Innovation/contribution	Sector	Type of study	Analysis technique	Key findings
Sivathanu (2019)	UTAUT2, IRT	Digital payment systems usage during India's demonetization	Digital Payment Systems	Empirical study Considering 766 respondents	PLS-SEM	Behavioral Intention and Innovation Resistance significantly affect actual usage of digital payment systems. The relationship between Behavioral Intention and Actual Usage is moderated by consumers' stickiness to cash payments
Khoo et al., (2024)	UTAUT2, IRT	Mobile Sales-force Automation (SFA) deployment by Coca-Cola to grocery retailers in Malaysia	Grocery retail	Quantitative cross-sectional survey	PLS-SEM	The study helps organizations validate and predict the acceptance and use of mobile SFA technology by grocery retailers. It aids in re-thinking mobile SFA technology investment and deployment strategies, especially in the context of recent disruptions caused by the pandemic and increasing sales costs

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Author contributions

ZRM conceptualised the research idea. He wrote a Research methodology, Analysis and Discussion sections. SP conceptualized the research idea, and wrote the Theoretical background, Hypotheses development, and Conclusion sections. RS wrote the Introduction and Research Implication sections. Finally, AT worked on developing survey instruments and data collection. All the authors reviewed and edited the manuscript.

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Competing interests

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